

# Four-Dimensional Integration: An Innovative Path to Cultivate Sequential Thinking in Senior Kindergarten Children Using Bee-Bot Programming Robots from the TPACK Perspective

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**Abstract.** The TPACK model provides an integrated perspective of technology, pedagogy and content knowledge for programming education in kindergartens. Integrating the TPACK perspective into Bee-Bot programming robot activities enables teachers to observe and respond to the development of children's sequential thinking, create a free exploration programming environment for children, and empower the growth of their logical sequencing abilities. Taking "Bee-Bot Treasure Hunt" as an example, teachers guide children through four stages – situational friction, independent planning, tool exploration, and experience creation. In each stage, children engage in multiple exploratory programming attempts, from which they distill one key insight about sequential thinking. This process continuously strengthens the sequential thinking ability of senior kindergarten children.

*ACM CCS (2012) Classification:*

Social and professional topics → Professional topics → Computing education → K-12 education

Social and professional topics → Computing education → Computational thinking

*Keywords:* TPACK perspective; Bee-Bot programming robots; senior kindergarten children; sequential thinking; N+1 practice

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## 1. Introduction

Mr. Tao Xingzhi once said: "Teach living students with living things anytime and anywhere." Vivid programming activity content originates from sequential situations and exploratory interests in children's daily lives. Through observation of programming activities for senior kindergarten children, it is not difficult to find that teachers often have problems such as disconnection between technical tools and teaching content, and overly abstract instruction explanations, which limit children's perception and construction of sequential logic. Integrating the TPACK perspective into Bee-Bot programming activities can prompt children to discover sequential clues in life, co-draw programming routes in independent planning, obtain iterative sequential thinking in tool exploration, and consolidate logical transfer ability in experience creation.

Early childhood programming education has become an important part of cultivating children's computational thinking in the 21st century. However, most existing practices focus on technical operation training rather than the cultivation of core thinking abilities such as sequential thinking. The TPACK model, which integrates technological pedagogical content knowledge, provides a theoretical framework for solving the problem of disconnection between technology and teaching in early childhood programming education. This study takes Bee-Bot programming robots as the carrier and constructs a "four-dimensional integration" N+1 practical path to explore how to effectively cultivate the sequential thinking ability of senior kindergarten children.

## 2. Methodology

This study adopts an action research method, taking 32 senior kindergarten children aged 5-6 years old in a kindergarten as the research objects, and carries out an 8-week "Bee-Bot Treasure Hunt" themed programming activity. Based on the TPACK perspective, the research constructs a four-stage cultivation path of situational friction, independent planning, tool exploration and experience creation. In each stage, children are allowed to carry out N types of independent explorations, and teachers guide them to condense 1 core experience of sequential thinking, forming a spiral upward development model.

## 2.1 Situational "Friction" to Arouse Interest: Collective Interaction to Anchor Sequential "Clues"

Mr. Tao Xingzhi believed that "Education takes effect immediately when life rubs against life." Children's perception of sequential thinking originates from the "friction" between life situations and game experiences. Teachers observe and record in daily life, ignite children's attention to sequence in collective interaction, anchor sequential "clues", and dynamically generate programming themes.

### 2.1.1 Free Activities: Recording "Sequential Moments" with Micro-videos to Discover True Problems

In free activities, children often spontaneously show naive cognition of sequence, such as "wash hands first, then eat" and "climb up first, then slide down". Teachers use mobile phones to record these "sequential moments" and edit them into micro-videos, which not only retain children's true expressions but also provide vivid materials for subsequent discussions.

#### Game Case 1:

During outdoor play, children lined up to play on the slide. Haohao said: "We have to take turns, slide one by one, no pushing." The teacher recorded this process with a mobile phone and shot a slow-motion replay of the children sliding down one by one. After the free activity that day, the teacher projected the video on the screen. The children watched with great interest and spontaneously imitated the slow motions in the video, gesturing and saying "him first, then me, then her". Haohao stood up and added: "When sliding on the slide, whoever is in front slides first, no cutting in line." The teacher asked: "What will happen if we don't follow the order?" The children chattered: "We will bump into each other" "We will cry". The teacher nodded with a smile and wrote the word "sequence" in the corner of the whiteboard.

### 2.1.2 Video Playback: Seminar via "Sharing Sessions" to Clarify the "Sequence List"

Teachers play back micro-videos and hold children's "sharing sessions" to guide children to review what other "ordered things" exist in life. Children are encouraged to express themselves through language, movements and painting to form a "sequence list", and vote to select the most interesting programming theme.

#### Game Case 2:

After watching the "lining up for the slide" video, children gave examples one after another: "Brush teeth first, then wash face in the morning" "When building blocks, build the base first, then the roof" "Stories should start with the beginning first, then the end". The teacher

recorded these sequential experiences to form a list, and the children clapped their hands for each item mentioned. After voting, "Let Bee-Bot hunt for treasure in order" got the highest votes. Several children shouted excitedly: "I want to be the treasure hunt captain". The teacher took out the Bee-Bot model and asked: "Then what should Bee-Bot do first, then what, for the treasure hunt?" Xiaoyu immediately said: "Press the clear key first, then the direction, and finally press GO." The teacher gave a thumbs up, and the children were full of expectations for the upcoming programming activities.

## 2.2 Independent "Planning" to Show Ideas: Independent Expression to Guide Diagrammatic Programming

Mr. Tao Xingzhi said: "Prepare various opportunities for students so that they can organize themselves and develop the ability to manage themselves." Based on children's initial perception of sequence, teachers trigger children's positive awareness of planning programming routes, support children's independent "planning", and improve and formulate programming "guidance" in independent expression and painting representation.

### 2.2.1 Identifying "True Problems" in Experience: Brainstorming to Generate Ideas

When children operate Bee-Bot for the first time, they often encounter real problems such as "pressing the wrong order", "not reaching the target" and "forgetting steps". Teachers guide children to conduct brainstorming in groups, identify logical contradictions in sequences, and represent creative solutions through painting.

#### Game Case 3:

When trying to make Bee-Bot walk from the starting point to the "treasure" position for the first time, many children failed. Xiaoyu said: "I pressed forward, forward, right turn, but Bee-Bot still hit the wall." An'an said: "You forgot to press the clear key first!" The teacher organized a group discussion: "How can we remember the correct order?" Each group of children drew "instruction cards" - arranging steps with arrows and numbers, and some children also drew "thumbs up" on the cards to indicate the correct order. The teacher further asked: "What if there are too many steps to remember?" Xiaoyu thought for a while and said: "We can try step by step, walk one step and record one step." The teacher encouraged them to stick their cards on the wall and follow them when operating next time. The children ran to stick the cards one after another, muttering "clear first, then walk, then turn".

### 2.2.2 Forming "New Plans" in Planning: Co-drawing Blueprints to Inspire New Interests

Teachers organize "programming round table meetings", where children in each group sit together, share their own route ideas, jointly draw Bee-Bot's treasure hunt map and instruction sequence, and clarify the starting point, end point, obstacles and steps. This cultivates children's planning ability and sequential awareness.

Game Case 4:

At the round table meeting, children drew treasure hunt plans in groups:

- Group 1: Starting point → 2 steps forward → right turn → 1 step forward → reach the "gem".
- Group 2: Starting point → 1 step forward → left turn → 3 steps forward → bypass the "river" (blue cardboard) → reach the "gold coin".

At the same time, children raised a new question: "What if two people want to play at the same time?" After discussion, they decided to take turns operating, and each person would change after programming three times. Some children volunteered to use an hourglass for timing.

### 2.3 Tool "Exploration" to Promote Unity of Knowledge and Action: Collaborative Progress to Iterate Sequential "Trends"

Mr. Tao Xingzhi emphasized that "Practice is the beginning of knowledge, and knowledge is the completion of practice." Following children's programming exploration process and their perspective, teachers let children realize the "unity of knowledge and action" in tool "exploration", and iterate the "trend" of sequential thinking together with peers in the process of continuously overcoming sequential errors and solving problems.

#### 2.3.1 Experience Review: Representing Sequential Problems

Experience review is a phased review link of children's programming activities. Teachers provide operation photos and videos to guide children to review the sequential problems encountered together, and represent core difficulties through language and painting, cultivating logical thinking ability and achieving the effect of "practice is the beginning of knowledge".

Game Case 5:

After the first round of "Bee-Bot Treasure Hunt", the teacher played the children's operation videos and encouraged them to review the problems and hold a "programming error correction meeting".

During the discussion, some children pointed at the screen and said "Look, it hit again", which caused a burst of laughter, and everyone became more active in finding problems.

### 2.3.2 Strategy Discussion: Discussing Solutions Together

Strategy discussion refers to supporting children to discuss problems and think about strategies based on the problem list sorted out by children. Teachers encourage each group to "claim" a problem, discuss solutions in the group, try to explore in the venue, and propose the final sequence optimization plan.

Game Case 6:

Each group claimed a problem and tried to solve it:

- Group A made an "instruction sequence recording board", and stuck an instruction card on the board every time a key was pressed to prevent omission or wrong order.
- Group B pasted "arrow stickers" on the top of Bee-Bot, and observed the arrow direction every time turning to strengthen the concept of left and right.
- Group C practiced repeatedly with the Bee-Bot simulation software on the tablet computer, simulating first then practicing.
- Group D proposed the "step-by-step verification method": check the position of Bee-Bot every 2 instructions executed to correct errors in time.

After the trial, each group also compared whose method was faster in front of the blackboard, and the atmosphere was warm.

## 2.4 Experience "Creation" to Expand Ideas: Consolidating Transfer to Plan Innovative "Directions"

Mr. Tao Xingzhi said in *\*Creative Education for Children\**: "Education cannot create anything, but it can inspire and liberate children's creativity to engage in creative work." Therefore, teachers carry out programming experience review links to consolidate children's sequential experience, promote the transfer and expansion of their ideas, mobilize their creative thinking, and plan innovative "directions" together in review, sharing and collision to promote the sustainable development of children's sequential thinking.

#### 2.4.1 "Two-way" Review: Clarifying Old Knowledge and Expanding New Ideas

"Two-way" review connects the "sequence list" sorted out by children in the early stage, guides children to expand new thinking about sequence on the basis of clarifying programming experience and sharing with peers. This promotes children's experience iteration and realizes the closed loop of programming activities.

##### Game Case 7:

When reviewing the experience of "Bee-Bot Treasure Hunt", teachers encouraged children to use multiple representations: "My most successful sequence" and "My most wanted new challenge". Children sorted out their experience through painting, language, role-playing and other ways. When the teacher presented the original "sequence list" again, the children's enthusiasm was rekindled. They voted to expand a new direction - "Bee-Bot Express Station": children drew community maps, planned the optimal order of multiple delivery points, and some children even volunteered to deliver "takeout".

#### 2.4.2 "Cross-class" Invitation: Sharing Together and Perceiving Value

"Cross-class" invitation mainly refers to teachers encouraging children to promote programming activities to other classes, invite them to participate in games and share fun, explain sequential logic and operation methods, and run the "little teacher system" through children's game process, so that children can fully perceive the value of sequential thinking and enhance their sense of achievement.

##### Game Case 8:

Children designed the "Bee-Bot Treasure Hunt Cheat Sheet" poster, and the teacher helped children record micro-videos of operation steps at the same time. Children took the posters and tablets to the middle class, carried out projection sharing and game explanation, and sincerely invited younger brothers and sisters to play Bee-Bot in the senior class. When the middle class children operated, the senior class children guided patiently: "Press the clear key first, then forward, right turn, forward." "Remember the order, don't press randomly." At the same time, children also absorbed the suggestion of middle class children to "add traffic light levels", and immediately made a traffic light with red cardboard and put it in the route after returning.

### 3. Results

After 8 weeks of action research, the "four-dimensional integration" N+1 path has achieved significant results in cultivating the sequential thinking ability of senior kindergarten children:

1. Interest and motivation improvement: 93.75% of children showed strong interest in programming activities, and could actively discover sequential clues in daily life.

2. Sequential logic ability enhancement: Children could accurately identify the three core elements of sequence (order, steps, direction), and could independently formulate complete programming routes from starting point to end point.

3. Problem-solving ability improvement: Children could classify programming problems independently and propose multiple solutions for the same problem, forming a positive attitude of "error correction = interesting challenge".

4. Experience transfer ability formation: Children could transfer sequential thinking to new scenarios such as "Bee-Bot Express Station", and could clearly express sequential logic in social interactions.

5. Social development promotion: Children developed good rules awareness and cooperation ability in group activities, and enhanced their self-confidence and expression ability through cross-class "little teacher" sharing.

### 4. Discussion

This study integrates the TPACK perspective with Tao Xingzhi's life education thought, constructing a "four-dimensional integration" N+1 path for cultivating sequential thinking in senior kindergarten children using Bee-Bot programming robots. The discussion below analyzes the underlying mechanisms, advantages over existing approaches, relationship to prior research, and study limitations.

#### 4.1 Underlying Mechanisms of the Four-Stage Path

The effectiveness of the four-stage path can be explained by three interlocking mechanisms.

First, embodied cognition in situated learning.\*\* In the situational friction stage, children encounter sequential logic through physical experiences — lining up, following routines, building blocks — rather than abstract instruction. According to embodied cognition theory, cognitive processes are grounded in sensorimotor experiences. This bodily understanding of



sequence transfers to programming commands, explaining why children in our study could later identify sequential clues in novel contexts without explicit prompting.

Second, cognitive scaffolding through external representation. In the planning stage, children offload sequential information onto physical artifacts—instruction cards, arrow sequences, treasure maps. This externalization reduces working memory load, which is particularly critical for 5 – 6-year-olds whose memory capacity is still developing. Tools like the "instruction sequence recording board" enable error detection that would otherwise be invisible, allowing children to monitor and revise their thinking in real time.

Third, iterative error-driven learning. In the tool exploration stage, children engage in "productive failure": encountering errors, diagnosing causes, testing solutions, and revising. This cycle—plan, execute, observe, diagnose, revise—parallels computational thinking's debugging process. The collaborative context amplifies this effect: when groups compare methods or explain strategies to peers, reasoning becomes externalized and collectively refined, fostering both individual skills and meta-cognitive awareness.

## 4.2 Advantages and Relationship to Previous Research

Compared to existing early childhood programming approaches, our path offers three distinctive advantages.

First, it shifts from technical operation to thinking cultivation. Many existing programming activities remain at the level of button-pressing without developing conceptual understanding of order. Unlike studies that emphasize teacher AI literacy or compare robots versus blocks, our study demonstrates a specific pedagogical sequence that systematically builds sequential thinking—from embodied experience to abstract planning to error-driven refinement to creative transfer.

Second, it operationalizes child-centered pedagogy within technological environments. A persistent challenge in early childhood technology integration is maintaining child agency while using structured technological tools. Our N+1 model resolves this tension: children drive multiple explorations (N), while teachers provide just-in-time guidance to distill core insights (1). This aligns with Vygotsky's zone of proximal development, where teacher intervention occurs precisely when children have sufficient experiential foundation to conceptualize their discoveries. This contrasts with curriculum designs that emphasize teacher-led instruction, and with models that do not specify how teacher guidance scales with child autonomy.

Third, it establishes a closed-loop developmental pathway. Many existing programming activities for young children are episodic, lacking cumulative progression. Our four-stage path creates a spiral curriculum: friction grounds the concept, planning externalizes it, exploration tests and refines it, and creation transfers it to new domains. This progression—concrete to abstract, guided to independent, single to multiple contexts—mirrors established developmental trajectories in cognitive psychology. Children's successful transfer to the "Bee-Bot Express Station" without re-teaching provides empirical support for this cumulative design.

The cross-class "little teacher" component extends Tao Xingzhi's original concept to programming education. Children's ability to explain sequential logic to younger peers—"Press clear first, then forward, right turn, forward"—demonstrates meta-cognitive mastery, as teaching requires deeper processing than execution.

#### 4.3 Limitations and Future Directions

Several limitations warrant acknowledgment. First, the sample was drawn from a single kindergarten, limiting generalizability. Future research should replicate this study across diverse settings, including rural kindergartens and those with limited technological resources. Second, the 8-week duration, while sufficient for initial changes, does not capture long-term sustainability; longitudinal studies are needed to examine whether gains persist and whether early sequential thinking training benefits later computational or mathematical reasoning. Third, our outcome measures were primarily observational; future studies could incorporate standardized assessments to enable precise comparisons. Fourth, the relative contribution of each mechanism—embodied cognition, external scaffolding, error-driven learning—remains unspecified; experimental designs isolating these components would provide stronger causal evidence. Finally, teacher preparation requirements were not systematically examined; future research should investigate the professional development needed for effective implementation of the TPACK-based four-stage path.

#### 4.4 Summary

The "four-dimensional integration" N+1 path provides a theoretically grounded and empirically supported approach to cultivating sequential thinking. Its effectiveness arises from embodied cognition, external scaffolding, and error-driven learning within a child-centered, teacher-facilitated framework. By moving beyond technical training, operationalizing child-centered pedagogy in technological contexts, and establishing a closed-loop developmental

pathway, this approach addresses gaps in prior research and offers a practical model for early childhood programming education.

## 5. Conclusion

Under the guidance of the TPACK perspective, this study constructs a "four-dimensional integration" N+1 programming thinking cultivation path for children, allowing children to germinate 1 programming interest point, formulate 1 sequence plan, obtain 1 logical thinking advancement, and generate 1 creative extension in the four stages of situational "friction", independent "planning", tool "exploration" and experience "creation". This path effectively creates a closed loop path for the development of children's sequential thinking, and provides a practical reference for early childhood programming education.

Early childhood programming education should return to the essence of thinking cultivation, take children's life experience as the foundation, and use appropriate technical tools to create a free and exploratory learning environment. Only in this way can we truly empower the growth of children's core literacy and lay a solid foundation for their lifelong learning and development.

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